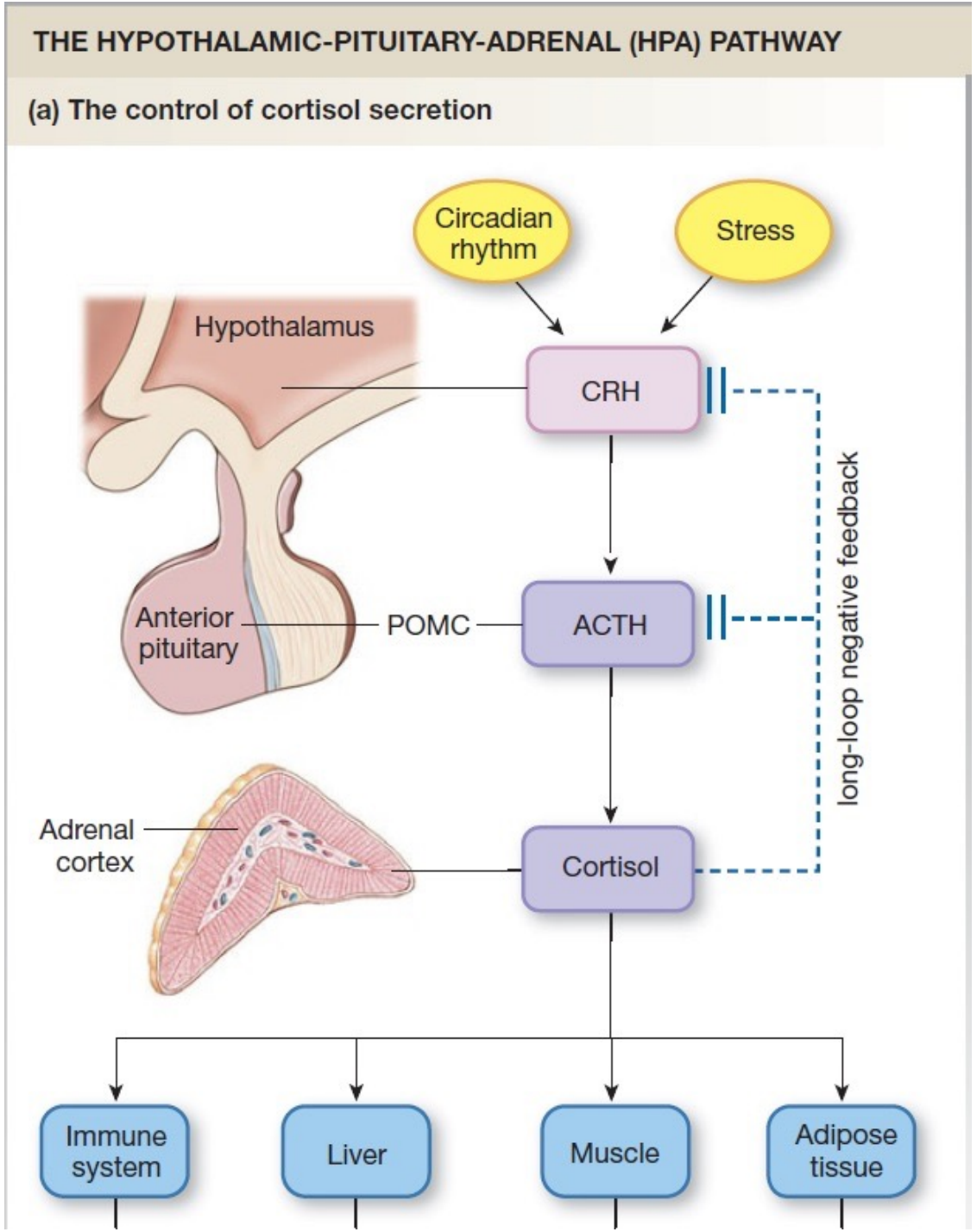
Pathophysiology of the Endocrine System

Pathophysiology of the Endocrine System

1. Introduction

Introduction

- Disorders are usually related to hyper or hyposecretion
- Disorders related to thyroid hormone, cortisol and growth hormone are usually due to changes in hypothalamic – pituitary axis
- Primary – damage is to the endocrine gland
- Secondary – damage is to the anterior pituitary or hypothalamus



Introduction

UNKNOWN

DUMB DOC

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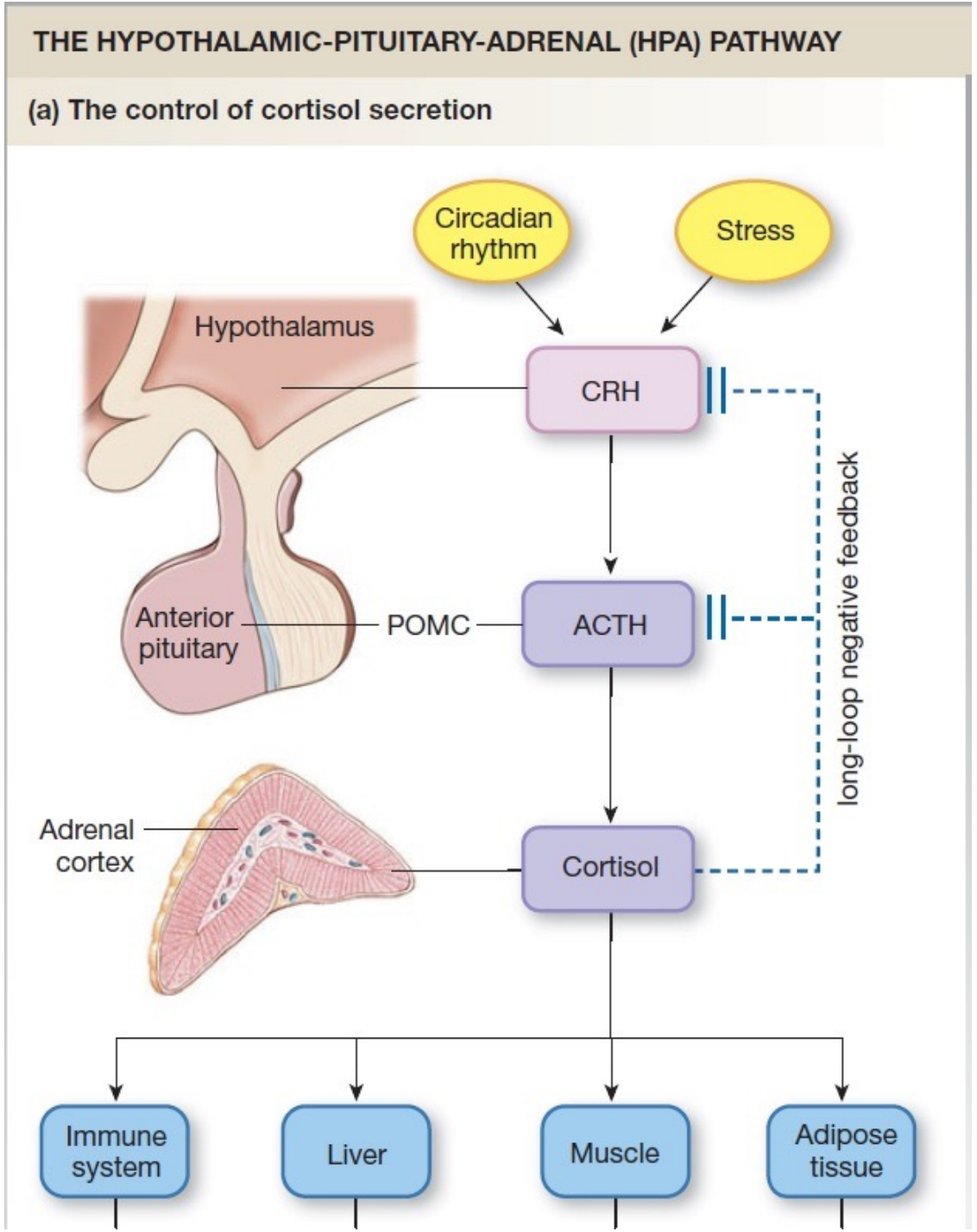
- Etiology
  - Idiopathic / iatrogenic / neoplastic
  - Congenital / autoimmune / infectious
  - Lack of precursors? Age-dependent critical periods?
- Pathophysiology
  - Depends on the cause but hormone secretion is altered
- Clinical Manifestations
  - Usually an exaggeration of normal effects of a hormone system
  - Usually very predictable



Pathophysiology of the Endocrine System

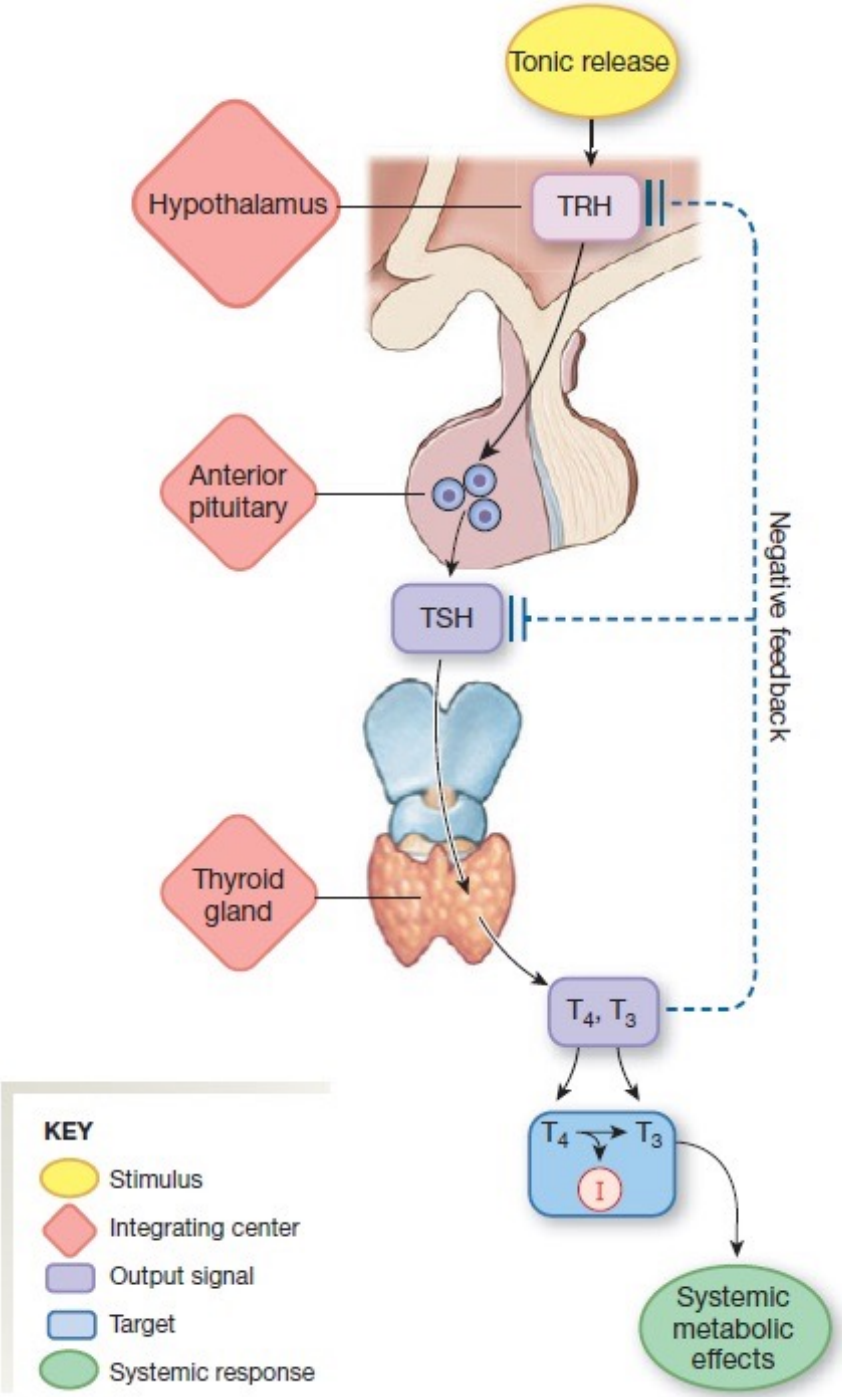
2. Disorders in the secretion of hormones of the thyroid gland

Effects of T3 & T4

- Increase basal metabolic rate (BMR; consumption of O2 and energy to meet minimum requirements to survive)
- Increases protein synthesis and protein degradation
- Increases rate of glycogenolysis and gluconeogenesis
- Increases lipolysis
- Increases HR, FoC and cardiac output

- In neonate, essential for normal neurologic development (particularly myelin, axon development & neurotransmitter production)

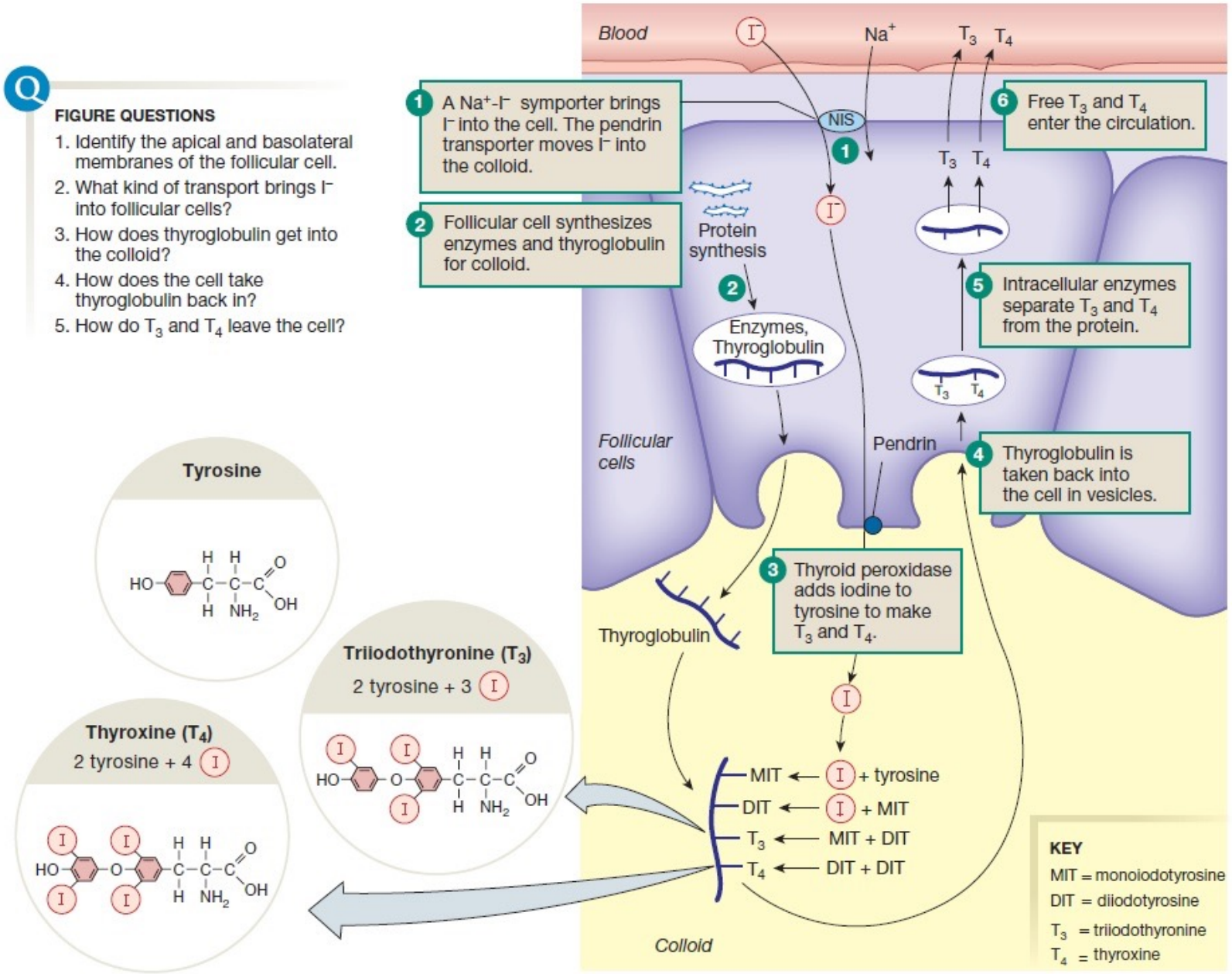
Regulation of Secretion of T3 & T4



Thyroid Hormones	
Cell of origin	Thyroid follicle cells
Chemical nature	Iodinated amine
Biosynthesis	From iodine and tyrosine. Formed and stored on thyroglobulin in follicle colloid.
Transport in the circulation	Bound to thyroxine-binding globulin and albumins
Half-life	6-7 days for thyroxine (T_4); about 1 day for triiodothyronine (T_3)
Factors affecting release	Tonic release
Control pathway	TRH (hypothalamus) → TSH (anterior pituitary) → T_3 + T_4 (thyroid) → T_4 deiodinates in tissues to form more T_3
Target cells or tissues	Most cells of the body
Target receptor	Nuclear receptor
Whole body or tissue reaction	↑ Oxygen consumption (thermogenesis). Protein catabolism in adults but anabolism in children. Normal development of nervous system
Action at cellular level	Increases activity of metabolic enzymes and Na^+-K^+-ATPase
Action at molecular level	Production of new enzymes
Feedback regulation	T_3 has negative feedback on anterior pituitary and hypothalamus.

Hypothyroidism – Etiology

- Congenital – thyroid dysgenesis
- Lack of precursors – non-iodinated salt
- Receptors on thyroid gland (TSH receptors) are insensitive to TSH
- Surgical removal / irradiation due to cancer



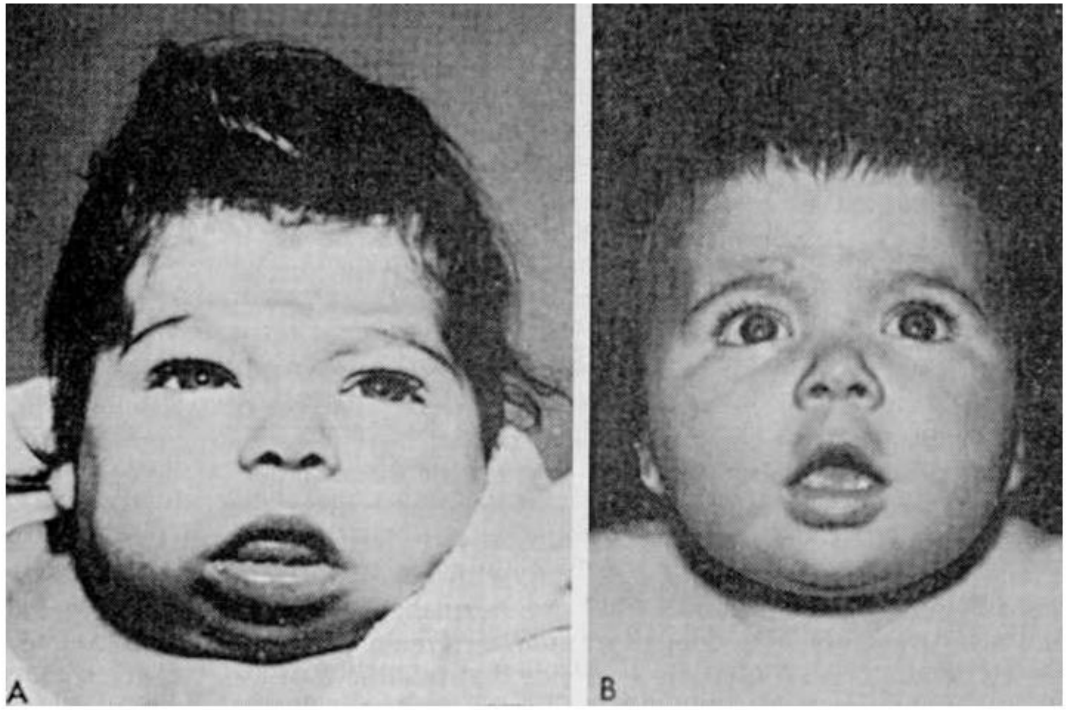
Generalized Clinical Manifestations of Hypothyroidism

- Decreased basal metabolic rate (BMR)
- Bradycardia, decreased blood pressure
- Weight gain despite decreased appetite
- Hyperlipidemia
- Constipation
- Cold intolerance; difficulty staying warm
- Lethargy / feeling of weakness
- Difficulty with memory & concentration; mentally slow
- Decreased T3 / T4 (if 1°) but increased TSH & TRH – why?

Hypothyroidism During Early Development

- T3 /T4 are critical for proper neurological development
- Cretinism
  - Signs appear at about 1 month of age
  - Thickened tongue & lips
  - Bradycardia
  - Poor muscle tone
  - Neonatal jaundice
  - Neurological delays
  - Irreversible damage if not treated
  - Managed with supplemental T3 /T4

6 MONTHS OLD BOY with CRETINISM & 4 MONTHS AFTER TREATMENT STARTED

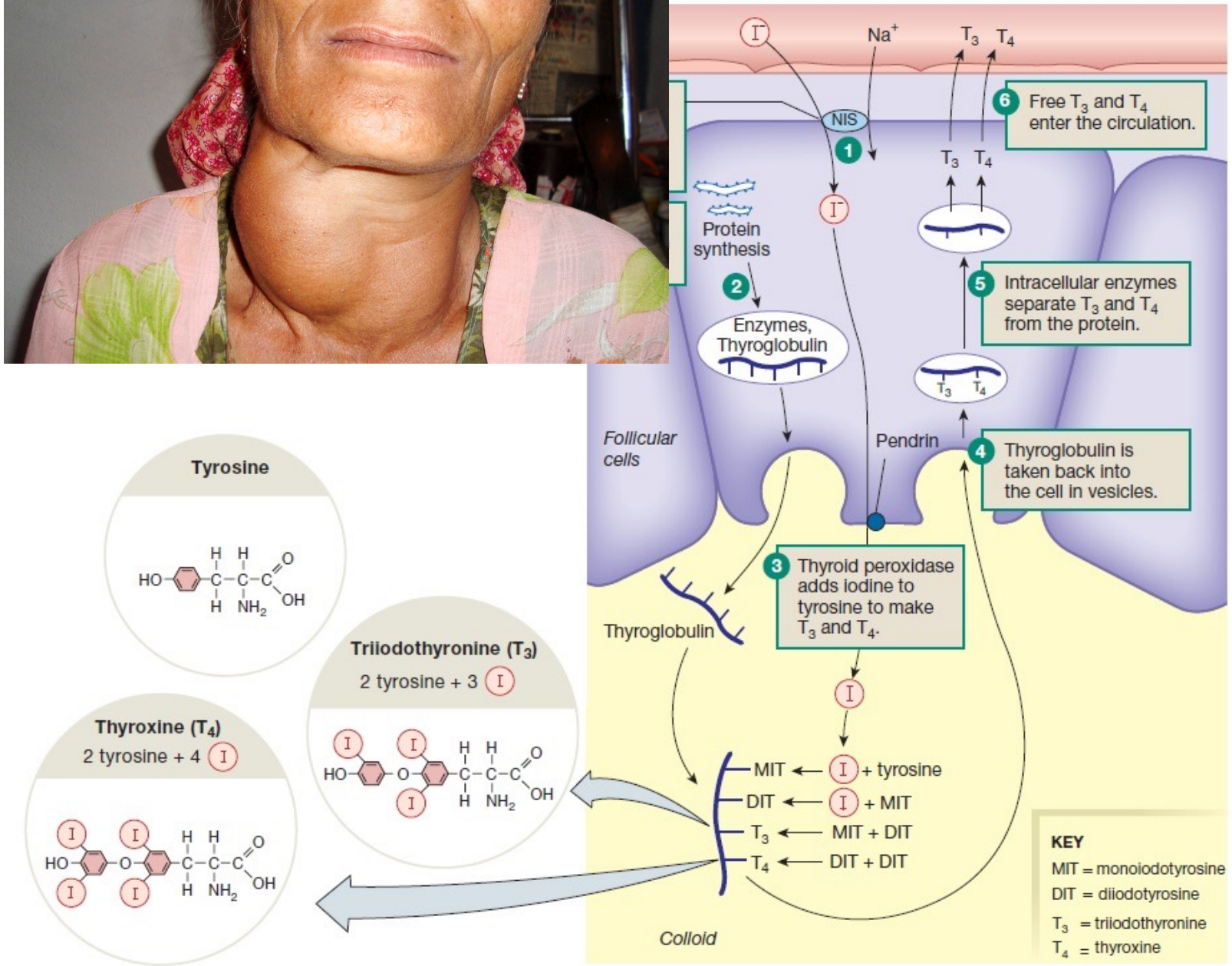


Hypothyroidism During Adulthood

- Myxedema (if not managed)
  - Prolonged lack of T3 / T4
  - Generalized non-pitting edema
  - Altered mental state; slow; flat
  - Cold intolerance / difficulty with maintaining body temperature
  - Managed with T3 / T4 supplements

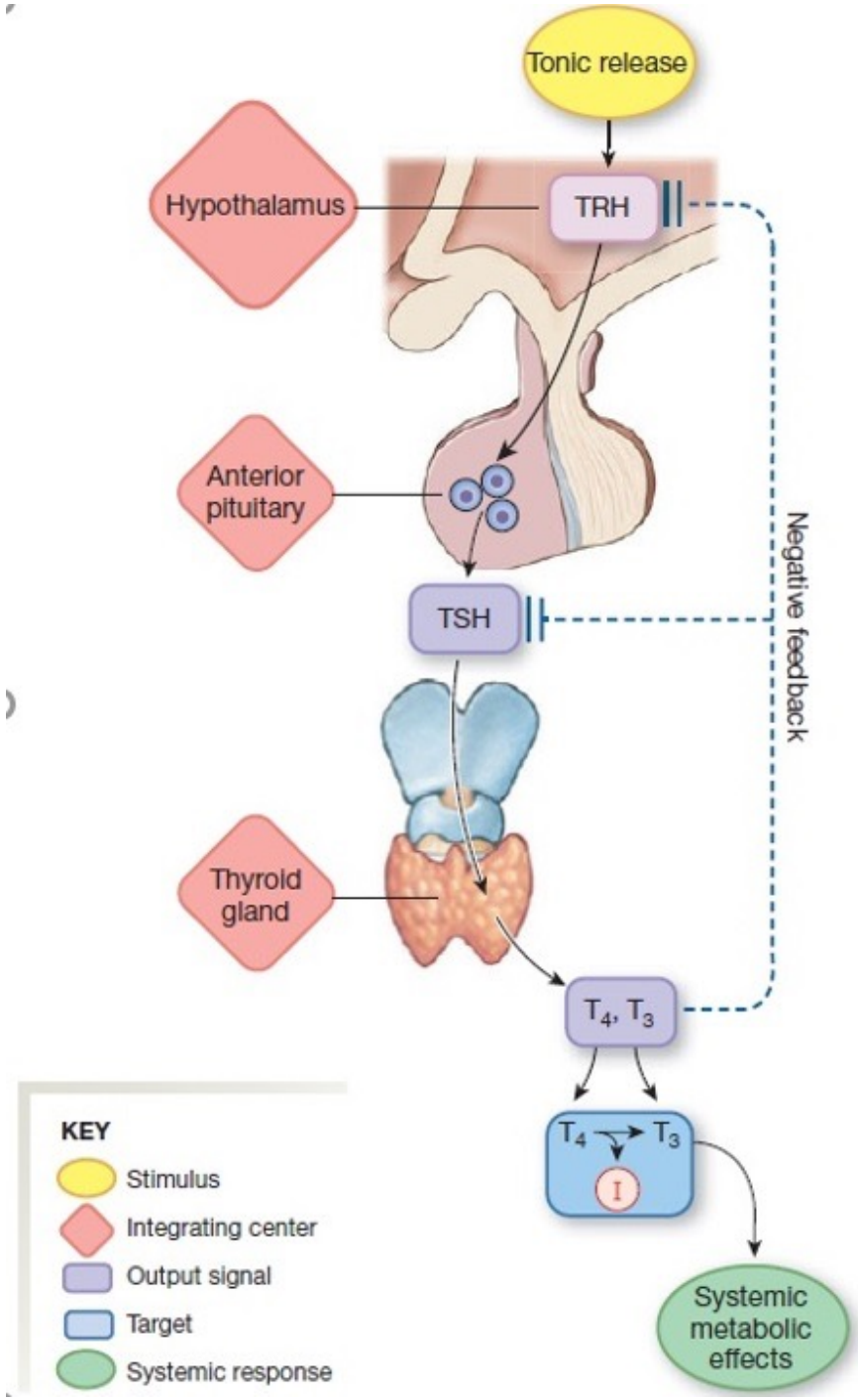
Hypothyroidism – lack of iodine

- Goiter develops
- TSH stimulates thyroid to produce T3 / T4
- No iodine so no T3 / T4 but thyroglobulin continues to be produced and enter colloid
- No T3/T4 thus TSH remains hgh
- Managed by adding iodide to the diet



Hyperthyroidism – Etiology

- Primary
  - Graves Disease
    - Antibody is produced that stimulates TSH receptors on follicular cells stimulating them to produce more T3 / T4
  - Thyroid gland tumour
- Secondary
  - Usually related to tumour of the anterior pituitary gland
  - How will this affect secretion of TRH & TSH?



Hyperthyroidism – General Clinical Manifestations

- Increased basal metabolic rate (BMR)
- Increased sympathetic nervous system activity
  - Tachycardia, diaphoresis
- Restlessness, insomnia, “spun”, hand tremors
- Weight loss despite increased appetite / N, V, D
- Heat intolerance
- Amenorrhea in females LACK OF MENSTRUATION
- Thyromegaly – due to constant production of T3 / T4
- Exopthalmos EYEBALL PROTUSION
- High T3 / T4; primary → almost no TRH, TSH – why?

Hyperthyroidism – Treatment

- Depends on the cause
- General
  - Beta blockers to address SNS related clinical manifestations
  - Drugs that inhibit T3 /T4 production
  - Removal of gland + T3 / T4 replacement therapy
  - Radioactive I-
- Secondary tumour
  - Removal of anterior pituitary

Pathophysiology of the Endocrine System

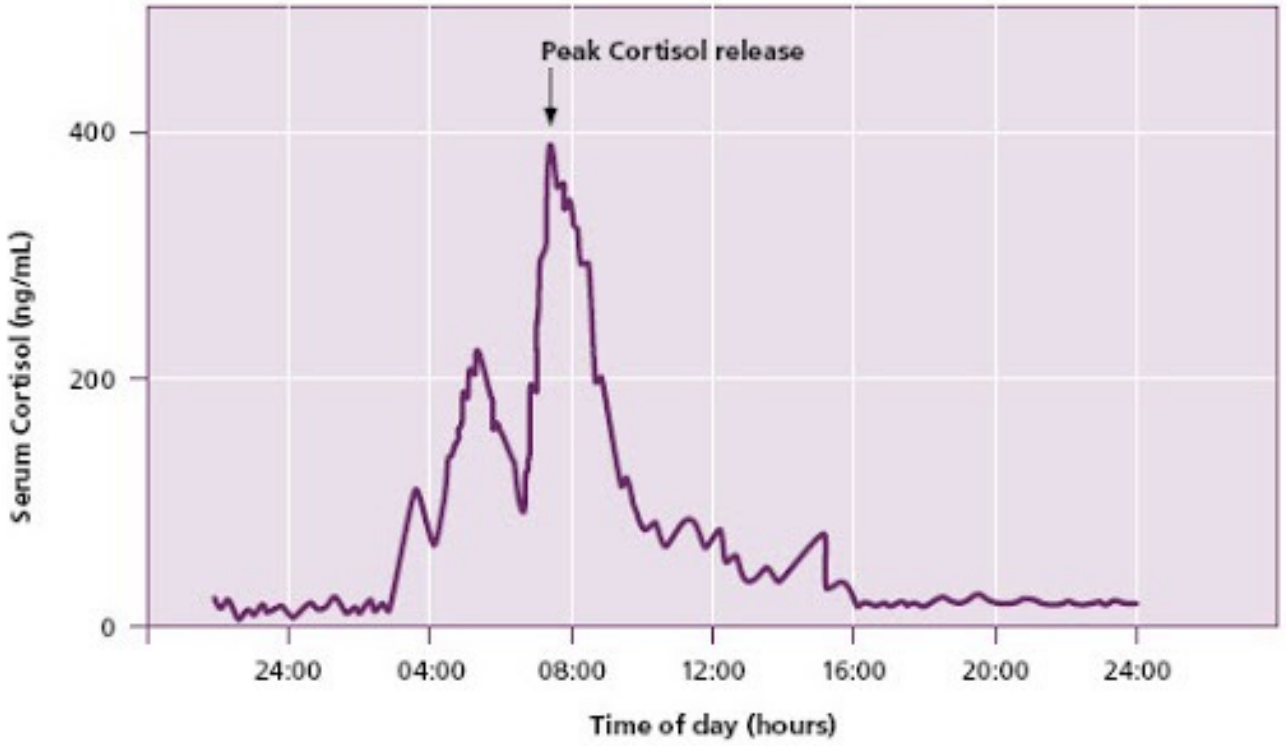
3. Disorders in the secretion of hormones by the adrenal cortex

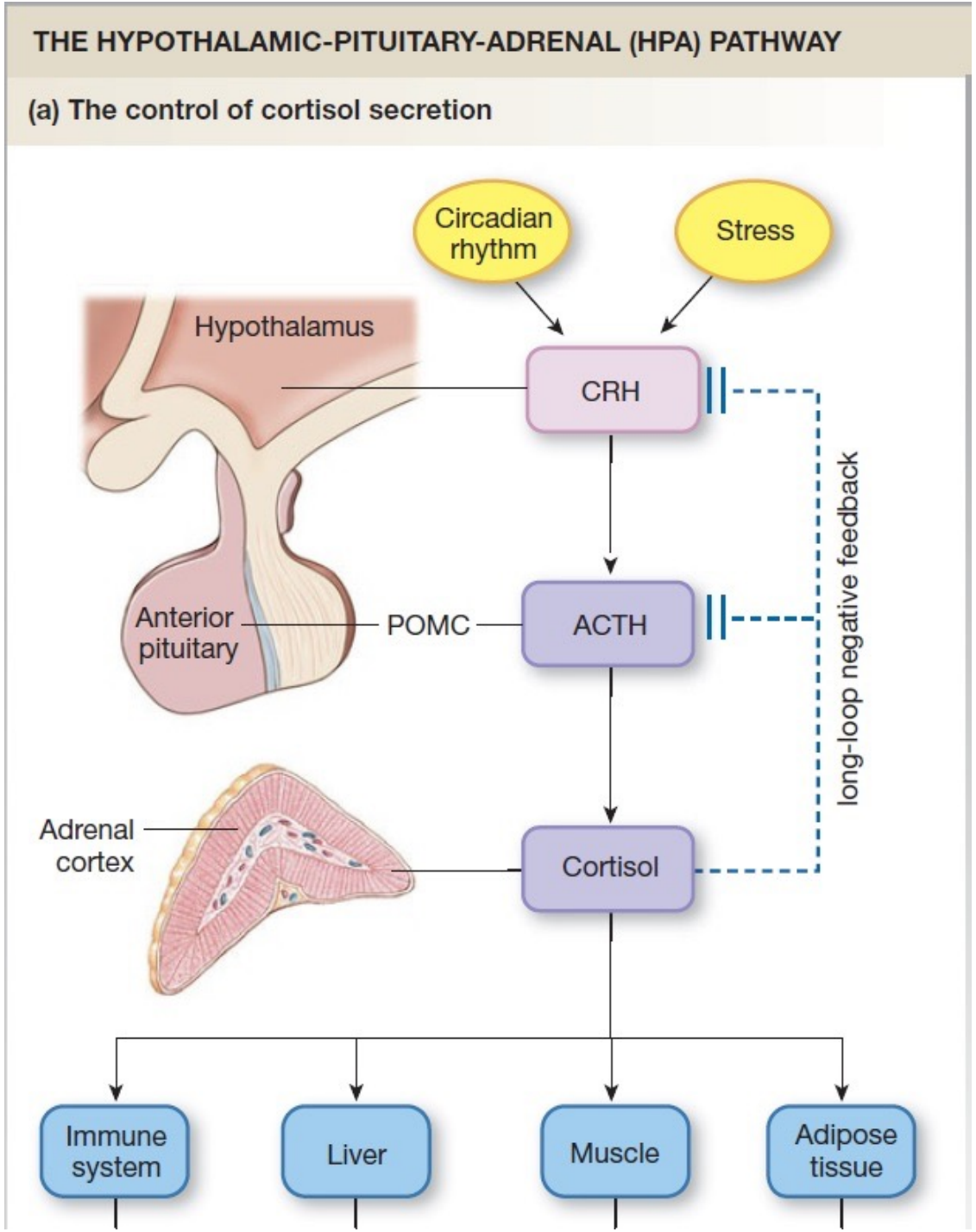
Glucocorticoid – Cortisol

- What does it do?
  - Increases blood glucose, AA and FA by altering metabolism of fats and proteins
  - Gluconeogenesis ← RELEVANT
  - Vasoconstriction (increases BP)
  - Anti-inflammatory
  - Stress hormone

Regulation Mechanisms

- Hypothalamic pituitary axis
- Circadian rhythms
- Stress





Hypocortisolemia

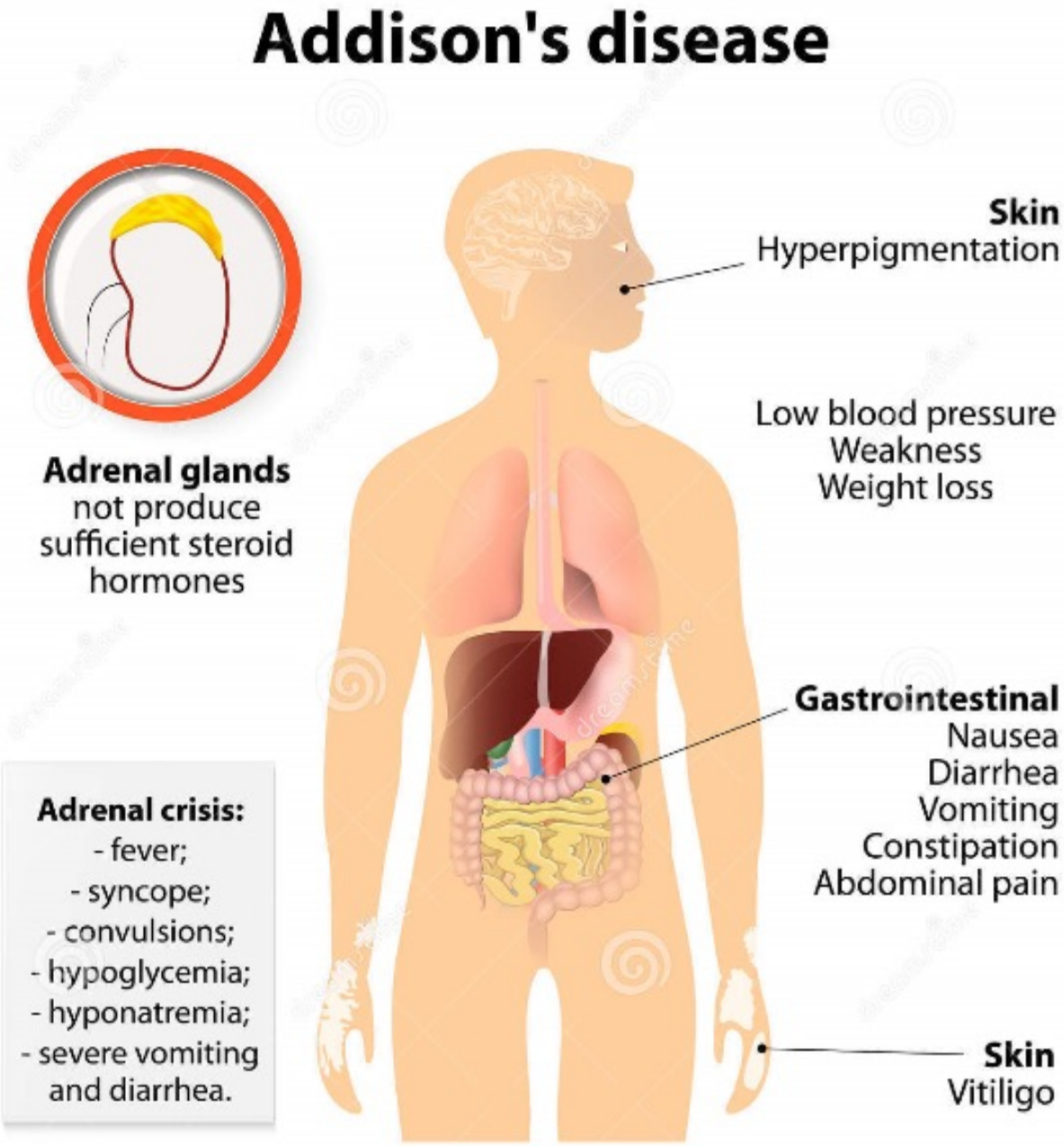
Caused by healthcare

- Etiology
  - Primary – Addison's disease – decreased production of cortisol and aldosterone
  - Secondary – due to reduced output of anterior pituitary (decreased ACTH)
  - Iatrogenic – prolonged exogenous cortisol or related steroid
- Pathophysiology (Addison's disease) / Clinical Manifestations
  - Too little aldosterone
    - Hyponatremia and hypovolemia
    - Hyperkalemia
  - Too little cortisol Adrenal Crisis/Addisonian Crisis
    - Decreased blood glucose
    - GI – N V D C
    - Increased pigmentation of skin – related to increased ACTH (POMC precursor) – but can sometimes lead to vitiligo (autoimmune Addison's disease)

and hypoglycemia

Addison’s Disease - Treatment

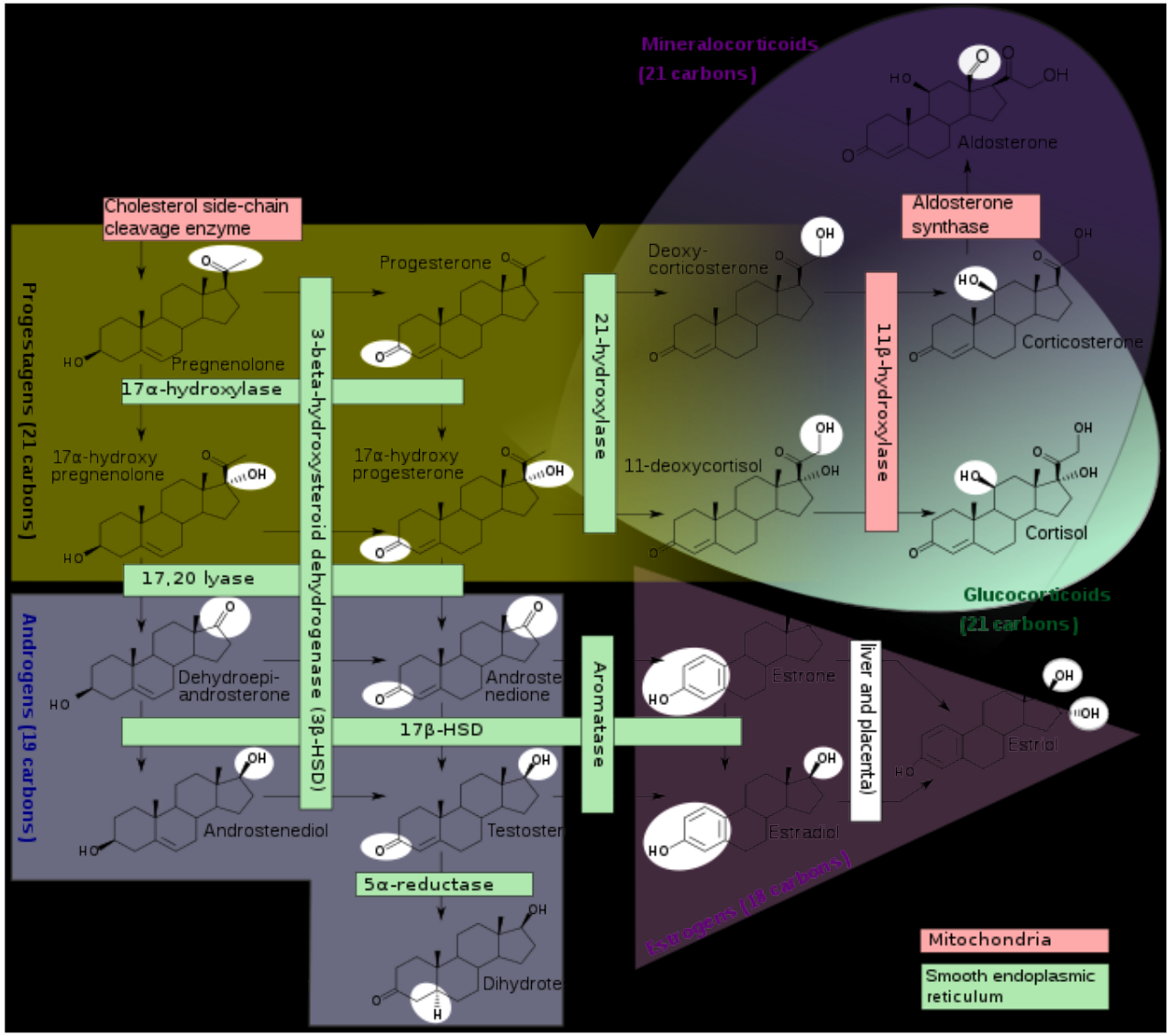
- Hormone replacement therapy with aldosterone and cortisol as required



https://healedbybacon.files.wordpress.com/2015/11/addisons.png?w=607&h=661

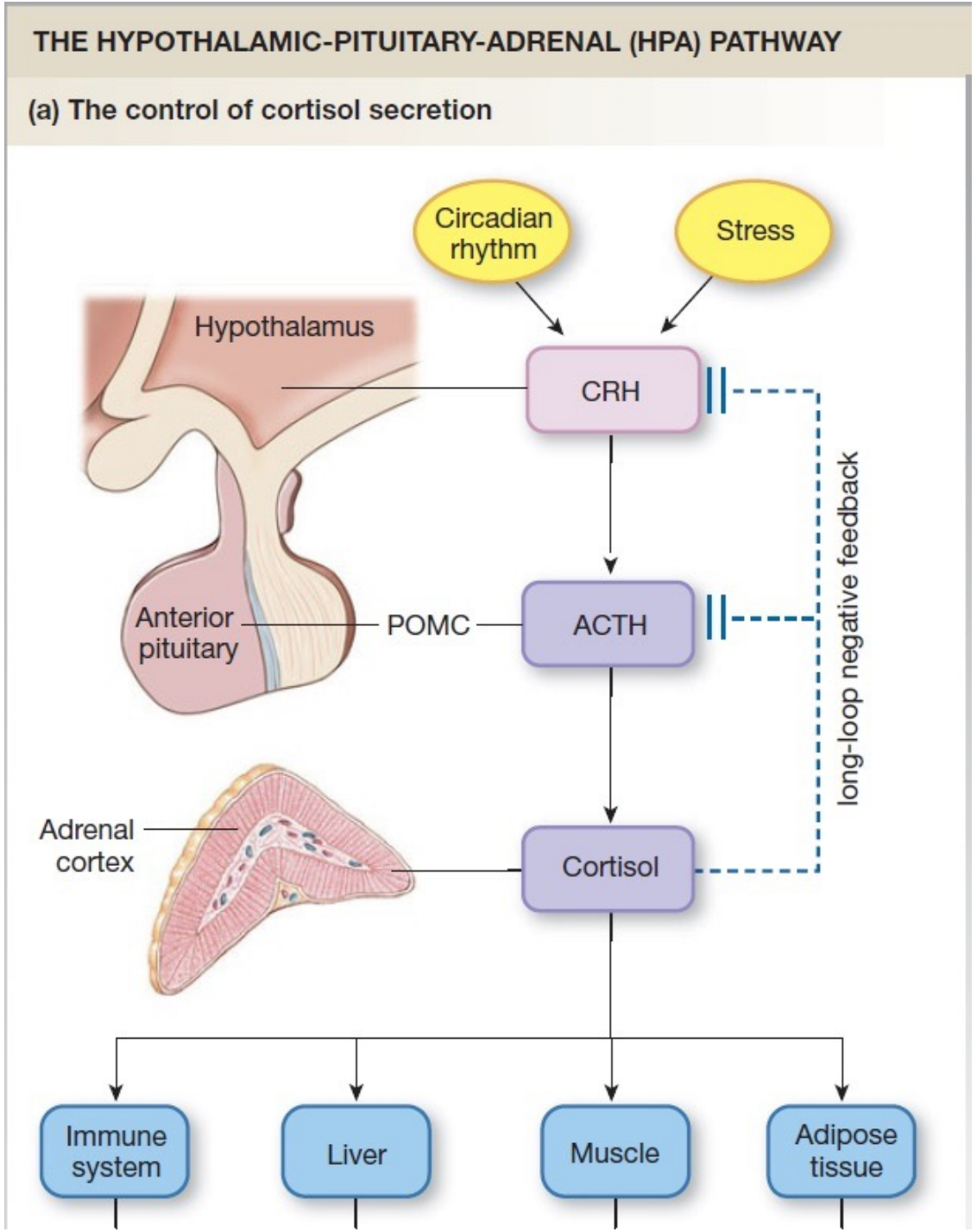
Congenital Adrenal Hyperplasia (Adrenogenital Syndrome)

- Etiology
  - Autosomal recessive condition; can't produce 21-hydroxylase, an enzyme necessary for the production of cortisol
- Pathophysiology
  - Causes increase production of androgens
  - Lack of cortisol → ↑ CRH & ACTH
- Clinical Manifestations
  - Virilization of females; enhanced maleness
- Treatment
  - Replacement therapy with cortisol



Cushing's Syndrome / Disease

- Etiology
  - Disease – primary or secondary → increased cortisol secretion; onset around age 40 depending on cause
  - Syndrome – iatrogenic
- Pathophysiology
  - In either case, hypercortisolemia can impact secretion of CRH and ACTH
  - Cushing's syndrome can sometimes completely stop endogenous production of CRH and ACTH
    - Pattern of delivery is usually on for a short period of time and then off for a period of time



Cushing's

- Clinical Manifestations
  - Fat depositions – buffalo hump, moon face, abdomen, weight gain
  - Decreased muscle mass; muscle weakness
  - Bone demineralization
  - Aldosterone –like effects
  - Increased production of androgens
  - Hyperglycemia (loss of sensitivity to insulin
  - Poor concentration and memory
- Treatment
  - Reduce exogenous
  - Deal with primary problem

Before and after onset of Cushing's disease

(a) Before (b) After

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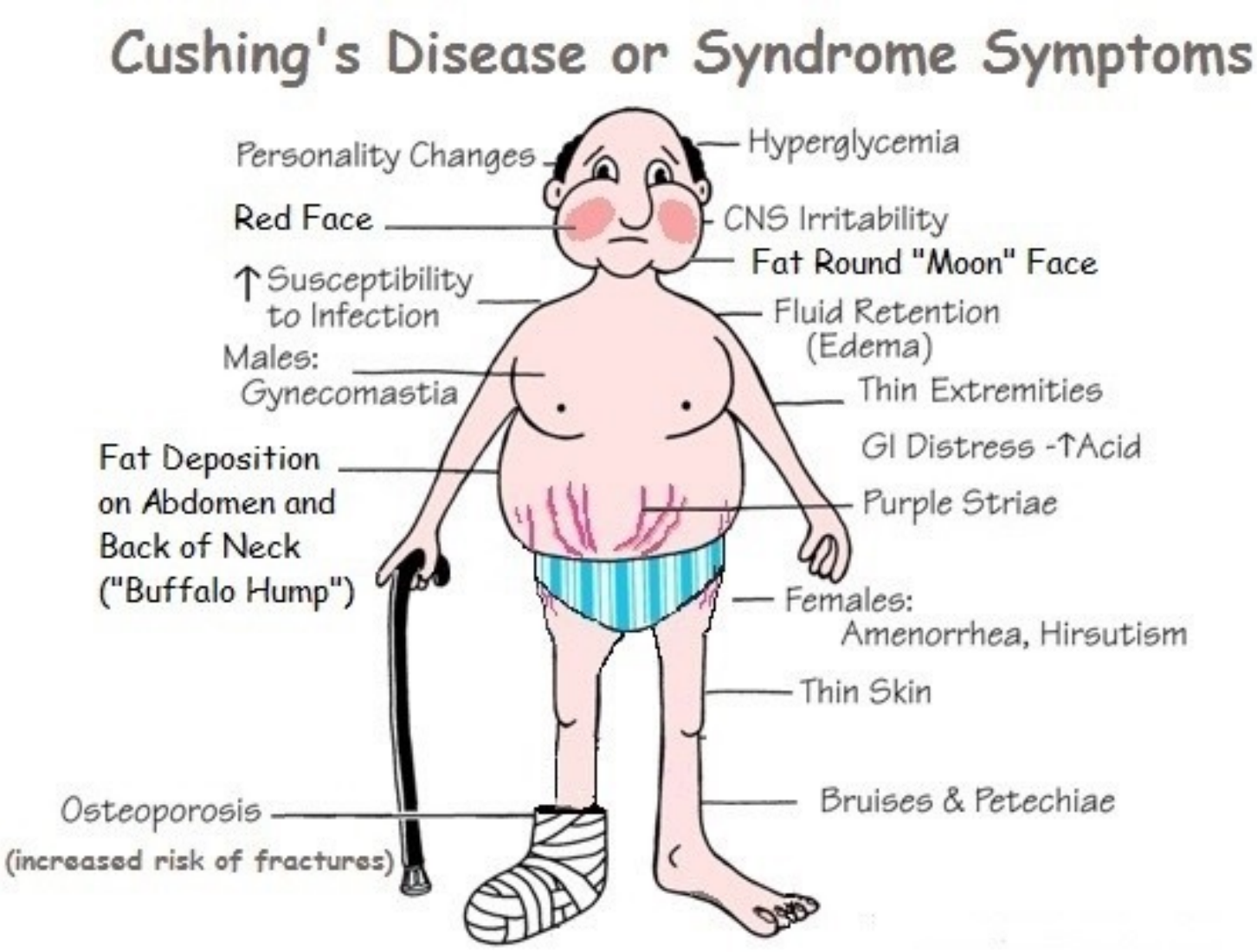
Cushing's



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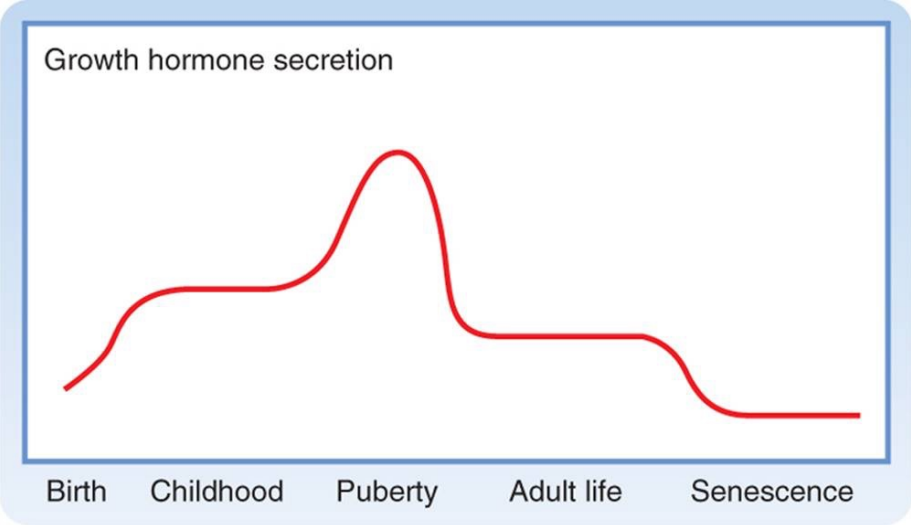
4. Disorders in the secretion of growth hormone

Growth Hormone

- Growth Hormone is produced by anterior pituitary
- GHRH promotes secretion; IGF negative feedback to anterior pituitary and hypothalamus reduces GH and GHRH secretion
- Somatostatin reduces secretion of GH
- GH can act directly on target tissues promoting bone and tissue growth or indirectly through IGF to also promote cartilage growth



Koeppen & Stanton: Berne and Levy Physiology, 6th Edition.
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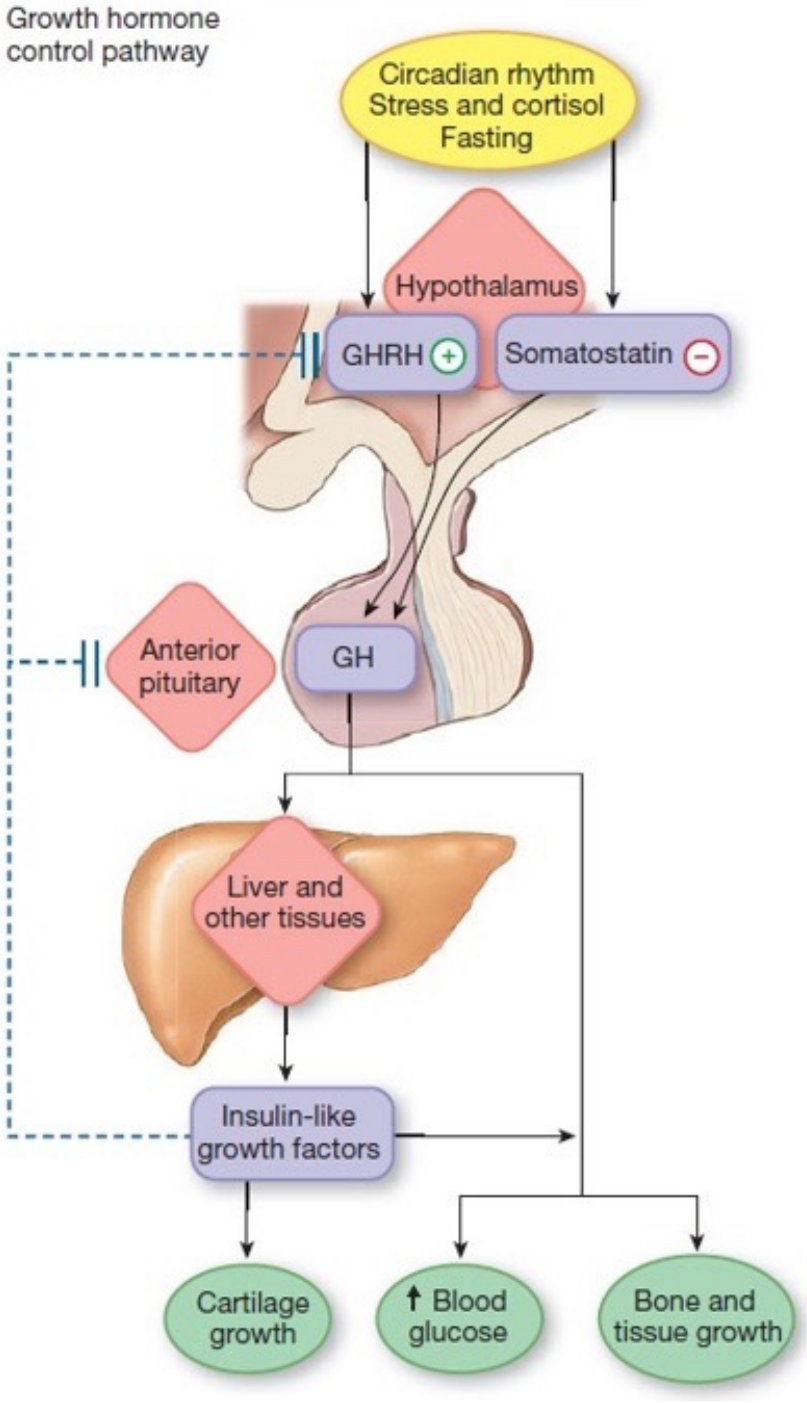

Hyposecretion of GH

- Etiology
  - Usually a congenital condition but can be acquired
- Pathophysiology
  - Lack of GH production; GH receptors don’t respond; lack of IGF production
- Clinical Manifestations
  - Normal birthweight / length
  - Reduced growth of long bones; normal proportions
  - Shorter axial skeleton; poor joint mobility
  - Hypoglycemia
  - Increased fat mass, decreased muscle mass
- Treatment – hormone replacement

Hypersecretion of GH

- Gigantism
  - Etiology
    - Pituitary gland tumour (benign) before epiphyseal plates stop growth
  - Pathophysiology
    - Increased GH & IGF secretion → increased bone growth and cartilage increased
  - Clinical Manifestations
    - Extremely tall individuals
    - Delayed puberty
    - Excessive bone deposition in face and hands
    - Cardiomegaly and CV disorders
  - Treatment
    - Removal of tumour

Hypersecretion of GH

- Acromegaly
  - Etiology
    - Pituitary gland tumour (benign) after epiphyseal plates stop growth; onset between age 20 and 40
  - Pathophysiology
    - Increased GH & IGF secretion → increased bone growth and cartilage increased
  - Clinical Manifestations
    - Excessive bone deposition in face and hands
    - Overgrowth of mandible → underbite
    - Arthritis, peripheral neuropathies
    - Cardiomegaly and CV disorders
  - Treatment
    - Removal of tumour